

NCBA Country Info Service — API Documentation

REST endpoint reference

NCBA Integration Microservices

API Documentation

REST API for the **Country Info integration microservice**. Given a country name, the service resolves it through the CountryInfo SOAP API, persists the result to MySQL, and serves CRUD operations over the stored data.

- **Base URL (local/Docker):** `http://localhost:8080`
- **Media type:** `application/json`
- **Resource identifier:** an opaque **UUID** (`id`). The internal sequential database key is never exposed — this prevents IDOR / record-enumeration attacks. Always use the `id` returned by a previous response.

Endpoint summary

#	Method	Path	Description	Success
1	POST	<code>/api/countries</code>	Ingest a country by name (via SOAP) and persist it	201 Created
2	GET	<code>/api/countries</code>	Fetch all countries	200 OK
3	GET	<code>/api/countries/{id}</code>	Fetch a country by its UUID	200 OK
4	PUT	<code>/api/countries/{id}</code>	Update a country by its UUID	200 OK
5	DELETE	<code>/api/countries/{id}</code>	Delete a country by its UUID	204 No Content

Observability (Spring Boot Actuator):

Method	Path	Description
GET	/actuator/health	Overall health (DB, SOAP, circuit breaker)
GET	/actuator/health/readiness	Kubernetes readiness probe
GET	/actuator/health/liveness	Kubernetes liveness probe
GET	/actuator/metrics	Micrometer metrics
GET	/actuator/prometheus	Prometheus scrape endpoint
GET	/actuator/info	Build/app info

Data model

CountryResponse

Field	Type	Notes
id	string (UUID)	Opaque public identifier
name	string	Country name (sentence-cased)
isoCode	string	ISO country code (e.g. KE) — unique
capitalCity	string	
phoneCode	string	International dialling code
continentCode	string	e.g. AF
currencyISOCode	string	e.g. KES
countryFlag	string (URL)	Flag image URL
createdAt	string (ISO-8601)	
updatedAt	string (ISO-8601)	
languages	array	List of { isoCode, name }

1. Ingest a country — POST /api/countries

Resolves the country name to an ISO code and full detail via SOAP, then **upserts** the record (keyed on ISO code: updates if it already exists, otherwise inserts). The name is converted to sentence case

server-side, so "kenya", "KENYA", and "kEnYa" all resolve to Kenya .

Request body

```
{ "name": "kenya" }
```

Field	Type	Required	Rule
name	string	yes	Must not be blank

curl

```
curl -i -X POST http://localhost:8080/api/countries \
-H 'Content-Type: application/json' \
-d '{"name":"kenya"}'
```

201 Created

```
{
  "id": "54bdedee-cb2e-4efa-955e-f355f4ba0cbf",
  "name": "Kenya",
  "isoCode": "KE",
  "capitalCity": "Nairobi",
  "phoneCode": "254",
  "continentCode": "AF",
  "currencyISOCode": "KES",
  "countryFlag": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Kenya.jpg",
  "createdAt": "2026-06-10T12:37:00.233",
  "updatedAt": "2026-06-10T12:37:00.233",
  "languages": [ { "isoCode": "swa", "name": "Swahili" } ]
}
```

Errors

Status	When
400 Bad Request	name is blank/missing
404 Not Found	The country name cannot be resolved to an ISO code
502 Bad Gateway	The SOAP service is unavailable / timed out / circuit breaker open

2. Fetch all countries — GET /api/countries

curl

```
curl http://localhost:8080/api/countries
```

200 OK

```
[
  {
    "id": "54bdedee-cb2e-4efa-955e-f355f4ba0cbf",
    "name": "Kenya",
    "isoCode": "KE",
    "capitalCity": "Nairobi",
    "phoneCode": "254",
    "continentCode": "AF",
    "currencyISOCode": "KES",
    "countryFlag": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Kenya.jpg",
    "createdAt": "2026-06-10T12:37:00.233",
    "updatedAt": "2026-06-10T12:37:00.233",
    "languages": [ { "isoCode": "swa", "name": "Swahili" } ]
  }
]
```

Returns `[]` when no records exist.

3. Fetch a country by id — GET /api/countries/{id}

`{id}` is the opaque UUID from a previous response.

curl

```
curl http://localhost:8080/api/countries/54bdedee-cb2e-4efa-955e-f355f4ba0cbf
```

200 OK — same shape as a single `CountryResponse` (see §1).

Errors

Status	When
404 Not Found	No record with that id

4. Update a country — PUT /api/countries/{id}

Updates the mutable fields of an existing record. The ISO code is the natural key and is not editable here. `name` is sentence-cased server-side. If `languages` is provided, it **replaces** the existing list.

Request body

```
{
  "name": "republic of kenya",
  "capitalCity": "Nairobi City",
  "phoneCode": "254",
  "continentCode": "AF",
  "currencyISOCode": "KES",
  "countryFlag": "http://www.oorsprong.org/WebSamples.CountryInfo/Flags/Kenya.jpg",
  "languages": [
    { "isoCode": "swa", "name": "Swahili" },
    { "isoCode": "eng", "name": "English" }
  ]
}
```

Field	Type	Required	Rule
<code>name</code>	string	yes	Must not be blank
<code>capitalCity</code> , <code>phoneCode</code> , <code>continentCode</code> , <code>currencyISOCode</code> , <code>countryFlag</code>	string	no	
<code>languages</code>	array	no	Replaces existing languages when present

`curl`

```
curl -i -X PUT http://localhost:8080/api/countries/54bdedee-cb2e-4efa-955e-f355f4ba0cbf \
-H 'Content-Type: application/json' \
-d '{"name":"republic of kenya","capitalCity":"Nairobi City"}
```

200 OK

```
{
  "id": "54bdedee-cb2e-4efa-955e-f355f4ba0cbf",
  "name": "Republic Of Kenya",
  "isoCode": "KE",
  "capitalCity": "Nairobi City",
  "...": "...",
  "updatedAt": "2026-06-10T12:40:11.512"
}
```

Errors

Status	When
400 Bad Request	name is blank/missing
404 Not Found	No record with that id

5. Delete a country — DELETE /api/countries/{id}

curl

```
curl -i -X DELETE http://localhost:8080/api/countries/54bdedee-cb2e-4efa-955e-f355f4ba0cbf
```

204 No Content — empty body on success.

Errors

Status	When
404 Not Found	No record with that id

Error response format

All handled errors return a consistent structured body (never a stack trace):

```
{
  "timestamp": "2026-06-10T10:00:00",
  "status": 404,
  "error": "Not Found",
  "message": "Country with id 54bdedee-cb2e-4efa-955e-f355f4ba0cbf not found",
  "path": "/api/countries/54bdedee-cb2e-4efa-955e-f355f4ba0cbf"
}
```

Validation (400) responses additionally include a `fieldErrors` array:

```
{
  "timestamp": "2026-06-10T10:00:00",
  "status": 400,
  "error": "Bad Request",
  "message": "Validation failed for request",
  "path": "/api/countries",
  "fieldErrors": [
    { "field": "name", "message": "name must not be blank" }
  ]
}
```

HTTP status codes used

Code	Meaning in this API
200 OK	Successful GET/PUT
201 Created	Country ingested/persisted
204 No Content	Successful DELETE
400 Bad Request	Request body validation failed
404 Not Found	Country/record not found
502 Bad Gateway	Downstream SOAP service failure / circuit breaker open
500 Internal Server Error	Unexpected error (safe generic message)

Postman

An importable collection and environment are provided under `postman/` :

- `postman/NCBA-CountryInfo.postman_collection.json`
- `postman/NCBA-CountryInfo.postman_environment.json`

The *Ingest country* request stores the returned UUID in a collection variable, so the Get/Update/Delete requests chain automatically.